Gemini AI made roadmap to help me achieve the goal of intercepting lethal harassment and high-toxicity communication before it reaches the victim, or in other words, stop cyberbullying.

Project Roadmap: Universal Safety Layer (USL)

Mission: To build an OS-level, privacy-preserving AI shield that intercepts lethal harassment and high-toxicity communication before it reaches the victim.

Green = Learned and completed

I. Foundations: Systems & Security (Months 1-4)

Goal: Understand the "terrain" where your code will live.

1. Low-Level Systems Programming

- **Language: Rust** (The industry standard for memory safety).
  - **Need to Know:** Ownership, borrowing, and the "Borrow Checker." This prevents the memory leaks hackers use to bypass security.
  - **Resource:** The Rust Programming Language (The Book)
- **Operating System Internals**
  - **Need to Know:** Kernel Space vs. User Space. You need to know how the OS manages processes to "hide" your layer from being turned off.
  - **Resource:** Linux Journey - Kernel Section

2. Networking & Interception

- **Protocols:** Master the **TCP/IP Stack** and **TLS/SSL Handshakes**.
  - **Need to Know:** How data is wrapped in "packets." Your USL acts as a filter for these packets.
  - **Resource:** Cisco Networking Essentials (Free)

II. The "Interceptor" Layer: Kernel & Hooks (Months 5-8)

Goal: Building the "hook" that catches messages in real-time.

1. Kernel Interception (eBPF)

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- **Technology: eBPF (Extended Berkeley Packet Filter)**
  - **Need to Know:** How to run sandboxed programs inside the OS kernel to monitor network traffic without crashing the system.
  - **Resource:** eBPF.io - Beginner's Guide

2. Windows & Mobile Drivers

- **Windows:** Study the **Windows Driver Kit (WDK)** for intercepting text at the system-input level.
- **Mobile:** Look into **Android Accessibility Services** (this is how apps "read" screens for safety).
  - **Resource:** Microsoft WDK Documentation

III. The "Brain": Privacy-Preserving AI (Months 9-12)

Goal: Making the AI smart enough to understand intent, but private enough to never leave the device.

1. Local AI Inference

- **Need to Know:** How to run "Small Language Models" (SLMs) on a device's CPU/GPU instead of a cloud server.
  - **Tools: ONNX Runtime** or **TensorFlow Lite**.
  - **Resource:** Hugging Face NLP Course

2. Advanced Privacy Techniques

- **Differential Privacy:** Adding "mathematical noise" to hide specific user data while keeping the "toxic" patterns detectable.
  - **Resource:** TensorFlow Privacy Guide
- **Federated Learning:** How to train the USL on many devices simultaneously without the data ever leaving those devices.
  - **Resource:** Google's Federated Learning Comic (Easy Start)

IV. Deployment & Impact (Year 1+)

Goal: Turning the project into a career/college asset.

1. Anti-Cheat & Gaming Integration

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- **Concept:** Study how **Easy Anti-Cheat (EAC)** operates. Your USL is essentially an "Anti-Cheat for Social Behavior."
- **Project:** Build a plugin for a game engine (Unity/Unreal) that uses your USL to mute toxic players automatically.

2. Competitions & Funding

- **CyberPatriot:** Join a team to prove your defensive skills.
- **MIT SOLVE:** Apply for the "Youth Safety" category.
  - **Link:** [MIT Solve Challenges]